

SYSTEM CALL

```
Activities Terminal Apr 19 14:56
tejini@localhost:~ — /usr/bin/vim pid.c

#include<stdio.h>
#include<unistd.h>
int main(){
    printf("PID: %d\n",getpid());
    printf("PPID: %d\n",getppid());
    return 0;
}
```

```
Activities Terminal Apr 19 14:55
tejini@localhost:~

[tejini@localhost ~]$ gcc pid.c
[tejini@localhost ~]$ ./a.out
PID: 3435
PPID: 2605
[tejini@localhost ~]$
```

```
Activities Terminal Apr 19 15:01
tejini@localhost:~ — /usr/bin/vim fork.c

#include<stdio.h>
#include<unistd.h>
int main(){
    if(fork()==0){
        printf("This is CHILD process\n");
    }else{
        printf("This is PARENT process\n");
    }
    printf("PID: %d\n", getpid());
    return 0;
}
```

```
Activities Terminal Apr 19 15:01
tejini@localhost:~

[tejini@localhost ~]$ gcc fork.c
[tejini@localhost ~]$ ./a.out
This is PARENT process
PID: 3494
This is CHILD process
PID: 3495
[tejini@localhost ~]$
```

```
Activities Terminal Apr 19 15:03
tejini@localhost:~ — /usr/bin/vim exec.c

#include<stdio.h>
#include<unistd.h>
int main()
{
    printf("Before exec()\n");
    execl("/bin/ls","ls",NULL);
    printf("After exec()\n");
    return 0;
}
```

```
Activities Terminal Apr 19 15:04
tejini@localhost:~

[tejini@localhost ~]$ vi exec.c
[tejini@localhost ~]$ gcc exec.c
[tejini@localhost ~]$ ./a.out
Before exec()
a a.out Desktop Downloads exec exec.c fork.c memory Music pc Pictures pid.c scheduling shellscrip sjf.c videos Videos
[tejini@localhost ~]$
```

```
Activities Terminal Apr 26 13:17
tejini@localhost:~ — /usr/bin/vim opendir.c

#include<stdio.h>
#include<dirent.h>
int main(){
    DIR *dir;
    dir = opendir(".");
    if(dir == NULL){
        printf("Failed to open directory\n");
        return 1;
    }
    printf("Directory opened successfully!\n");
    closedir(dir);
    return 0;
}
```

```
Activities Terminal Apr 26 13:17
tejini@localhost:~

[tejini@localhost ~]$ ls
a.out  Desktop  Downloads  exec.c  forl.c  Music  pc  pid.c  scheduling  sjf.c  test.sh  Videos
a.out  Documents  exec  fork.c  memory  opendir.c  Pictures  roundrobin.c  shellscript  tejini  videos

[tejini@localhost ~]$ vi opendir.c
[tejini@localhost ~]$ gcc opendir.c
[tejini@localhost ~]$ ./a.out
Directory opened successfully!
[tejini@localhost ~]$
```

```
Activities Terminal Apr 26 13:31
tejini@localhost:~/systemcalls — /usr/bin/vim readdir.c

#include<stdio.h>
#include<dirent.h>
int main(){
    DIR *dir;
    struct dirent *entry;
    dir = opendir(".");
    if(dir==NULL)
    {
        printf("Cannot open directory\n");
        return 1;
    }
    printf("Files in directory:\n");
    while((entry = readdir(dir)) != NULL)
    {
        printf("%s\n",entry->d_name);
    }
    closedir(dir);
    return 0;
}
```

```
Activities Terminal Apr 26 13:31
tejini@localhost:~/systemcalls

[tejini@localhost systemcalls]$ gcc readdir.c
[tejini@localhost systemcalls]$ ./a.out
Files in directory:
.
..
pid.c
fork.c
exec.c
opendir.c
readdir.c
a.out
[tejini@localhost systemcalls]$
```